

Adaptive Modulation and Coding Technology in 5G System

Yue Wang
Chongqing University of Posts and
Telecommunications
Chongqing, China
s170131126@stu.cqupt.edu.cn.

Wei Liu
Chongqing University of Posts and
Telecommunications
Chongqing, China
s170131126@stu.cqupt.edu.cn.

Linquan Fang
Yunnan Innovation Institute · BUAA
Kunming, China
fanglq1991@163.com.

Abstract—Internet of Things(IoT) makes everything intelligently possible. In the IoT era, the application of products such as smart cities and smart homes will provide people with a more convenient and quality life. 5G is the prelude and foundation for achieving this. In the 5G mobile communication system, the base station performs adaptive coding and modulation based on the channel state information fed back by user, and improves the spectrum efficiency by selecting different combinations of modulation types and code rates. Therefore, the channel state information estimated by receiver side needs to be mapped to a value measuring the overall channel state and fed back to the base station. This paper first simulates the modulation and coding schemes(MCS) given in the 5G standard under Gaussian channels, and obtains the SNR-BLER curves of different MCS. Then, the scheme of SNR mapping to CQI is improved, and a new mapping relationship between SNR and CQI is obtained by performing region fitting. Finally, for the adjustment factor optimization process, a new adjustment factor is introduced and the calculation formula for the optimization factor is modified. The BLER is taken as the logarithm and then used Logarithmic results to find the mean square error, thereby reducing mapping errors Compared with the traditional scheme, the proposed scheme has higher throughput, so as to better promote IoT technology.

Keywords—Internet of Things; Adaptive modulation and Coding; Signal-to-Noise Rate; Modulation and coding Scheme

I. INTRODUCTION

In recent years, IoT technology has been developed rapidly and widely. With the successive deployment and commercialization of 5G, the combination of Internet of Things and 5G technology has become the trend of technological development[1]. The Internet of Things is the basis for 5G networks and a platform for 5G value realization. In terms of the IoT field, the value brought by 5G is diverse and comprehensive. For example, the rapid popularity of high-definition video call services has caused the development of the live broadcast industry, smart homes and factories, virtual reality, and artificial intelligence-based personal assistants. Unprecedented technology and applications are also changing our lives and providing us with many conveniences. The communication mode has gradually changed from people-centric to material-centric. New services have higher requirements for data volume and transmission rates, leading new research directions for technology, and prompting another change in mobile communication technology. According to the schedule of the 5G technology standard issued by the International Telecommunications Union (ITU), the 5G technology plan will be completed in 2020 and successfully commercialized.

At present, researchers have done a lot of research on the key technologies for LTE / 5G, and have made great

progress[2][3]. The development direction of mobile communication technology is more and faster transmission data. Now, the LTE's peak rate can reach at 1Gbit / s, and peak rate of 5G can reach is 10Gbit / s, and in some cases may support up to 20Gbit / s[4]. However, a high data rate requires higher spectrum resources, and the existing frequency band resources are limited. This requires link adaptation technology to maximize spectrum utilization when frequency band resources are limited. The traditional schemes are designed with the assumption that the channel quality is very poor, which will cause a waste of spectrum resources [5]. With the increase in demand for mobile communications and the development of technology, how to improve the utilization of frequency bands has become a problem that must be solved. Adaptive Modulation and Coding (AMC) technology is an excellent way to solve this problem.

In the future, 5G mobile communication systems will still adopt adaptive coding and modulation technology as the main means to improve the frequency band utilization rate. The purpose of adaptive coding and modulation is to improve link spectrum utilization, reduce system bit error rate (BER), make transmission power constant, reduce interference to other users, or meet the different needs of various types of data[6]. Its basic principle is to adjust the coding and modulation methods according to the current channel quality changes within the limits of the system, and maximize the data transmission rate while ensuring the quality of data transmission. In an adaptive coding and modulation system, when the channel condition is excellent, a higher-order MCS is used (that is, a higher code rate and higher-order modulation are used, such as 64QAM, etc.), thereby improving transmission efficiency and spectrum utilization. When the channel quality is poor, the higher transmission rate will bring serious inter-symbol interference to the receiving end. At this time, a lower coding and modulation method (low code rate and low-order modulation type like QPSK) is needed. Although the transmission rate is reduced, the bit error rate(BER) at the receiving end is also reduced, thereby ensuring the accuracy of the transmitted data and ensuring the overall throughput. Finally, the goal of dynamically matching the MCS with the channel condition in various situations is achieved [7][8].

The latest research on adaptive modulation and coding technology is mainly focused on OFDM, Multi-Input Multiple-Output(MIMO) and other aspects. Combining AMC technology with OFDM technology, as a key technologies of LTE/5G, not only improves the flexibility of the system, but also enhances the reliability of the AMC system. MIMO is a multi-transmit antenna and multi-receive antenna system, which makes full use of space resources. The AMC technology based on the MIMO system can make

each transmission channel reach the maximum capacity, thereby effectively improving the data transmission rate[9][10]. Therefore, the combination of AMC technology and other technologies has become an important method to improve transmission efficiency. The mainstream cellular system standards have adopted AMC technology for wireless communication transmission. Many international organizations such as IEEE, 3GPP, etc. also adopt adaptive modulation and coding technology into wireless communication technology standards.

This paper mainly studies adaptive coding and modulation schemes in 5G systems. It is assumed that the modulation and coding schemes are adaptively selected on the premise of known channel state information. First, the modulation and coding schemes in the three tables in 38.214 are simulated under a Gaussian channel to obtain the SNR-BLER curve, and then the SNR threshold when BLER = 0.1 is obtained. Because the traditional CQI selection needs to store the relationship between SNR and CQI in the table, each time it is selected, the process is tedious. In this paper, the solution is improved by region curve fitting, and the corresponding expression of CQI and SNR is summarized.

II. ADAPTIVE MODULATION AND CODING

A. Principle of adaptive coding and modulation

AMC technology dynamically adjusts the modulation order, coding method and coding rate of symbols to maximize throughput of the entire link transmission system [11], because throughput of the OFDM system is related to the modulation method, coding method and bit error rate. Generally, we will set the target error rate of the system. For the reason of maximizing the system's transmission performance and throughput under a fixed BLER, we should dynamically adjust its modulation and coding method at the transmitting end so that it can The time-varying fading channel (such as a frequency-selective channel) adapts to the change of the channel, maintains the stability of the transmission of the system, as well as improves throughput. The visual description is that while the channel environment is better, we use a higher transmission rate to transmit data. If the channel environment is deficient, for the purpose of correctly receive the signal at the receiving end, a low transmission rate is used.

B. AMC implementation in 5G

The 5G physical layer downlink AMC process is: the signal is sequentially subjected to LDPC encoding, scrambling, modulation, layer mapping, and precoding at the transmitting end, and then the OFDM symbols are generated and mapped to different antenna ports, and then passed through the channel to the large receiving end; The client performs your processing process, first demodulating OFDM, and then performing channel estimation, resource mapping, demodulation, descrambling, and decoding.

The 5G physical layer downlink mainly uses LDPC coding, and the modulation methods include QPSK, 16QAM, 64QAM, and 256QAM. Table 1, Table 2 and Table 3 are the CQI tables defined by 3GPP TS38.214. This article mainly studies the MCS and its switching threshold in these three tables.

III. CQI SELECTION ALGORITHM

A. CQI selection Algorithm

The main steps of CQI feedback are three steps: the first is the generation of a mapping table between SNR and CQI; the second is the acquisition of channel state information. UE obtains the parameters of the current channel environment and feeds it back to the transmitter; the third is that the transmitter selects a transmission scheme according to the acquired CQI. The content of the transmission scheme in this study includes TB block size, modulation method, and code rate. The TB size is depending on the number of REs and MCS. The number of REs occupied by downlink transmission is fixed, so different transmission schemes here correspond to different modulation and coding schemes.

The specific implementation steps are as follows:

1, Simulate the transmission schemes (modulation methods, code rates) corresponding to the 15 CQIs shown in Tables 1, 2, and 3[12] in a Gaussian channel environment to obtain the relationship between the BLER and SNR graph.

2, According to the relationship curve diagram generated in the first step, sequentially compare the SNR threshold values corresponding to BLER = 0.1 in the relationship curves corresponding to each CQI.

3, When transmitting in an actual fading channel, calculate the effective SNR under the AWGN channel.

4, After the receiving end obtains the effective SNR, it queries the threshold table of different CQIs obtained in the second step, and maps the effective SNR value reflecting the current channel quality to the CQI value, and then feeds back CQI to the transmitting end in the form of CQI.

5, After receiving the CQI feedback from the terminal UE, the sending end selects a sending scheme corresponding to the CQI for downlink data transmission according to the received CQI lookup table.

TABLE I. 4-BIT CQI TABLE 1

CQI index	modulation	code rate *1024	efficiency
0		Out of range	
1	QPSK	78	0.1523
2	QPSK	120	0.2344
3	QPSK	193	0.3770
4	QPSK	308	0.6016
5	QPSK	449	0.8770
6	QPSK	602	1.1758
7	16QAM	378	1.4766
8	16QAM	490	1.9141
9	16QAM	616	2.4063
10	64QAM	466	2.7305
11	64QAM	567	3.3223
12	64QAM	666	3.9023
13	64QAM	772	4.5234
14	64QAM	873	5.1152

CQI index	modulation	code rate *1024	efficiency
15	64QAM	948	5.5574

TABLE II. 4-BIT CQI TABLE 2

CQI index	modulation	code rate *1024	efficiency
0		Out of range	
1	QPSK	78	0.1523
2	QPSK	193	0.3770
3	QPSK	449	0.8770
4	16QAM	378	1.4766
5	16QAM	490	1.9141
6	16QAM	616	2.4063
7	64QAM	466	2.7305
8	64QAM	567	3.3223
9	64QAM	666	3.9023
10	64QAM	772	4.5234
11	64QAM	873	5.1152
12	256QAM	711	5.5547
13	256QAM	797	6.2266
14	256QAM	885	6.9141
15	256QAM	948	7.4063

B. Effective SNR mapping Algorithm

Due to the frequency selection characteristics of wireless multipath channels, when the system bandwidth is large, the channels of each subcarrier may have different characteristics, resulting in large changes in the SNR. Therefore, UE needs to perform effective SNR mapping through a certain function, and map the SNRs of all time-frequency resource particles into an SNR that can reflect the overall channel state. The receiving end compares the obtained effective SNR SNR_{eff} with the SNR threshold when BLER = 0.1 is simulated under AWGN-SISO conditions, so as to select the highest CQI value when BLER < 0.1 is met[13].

The main objective of the effective SNR mapping algorithm is to establish a function that could reflect the polymorphic channel to an alternative single channel, and the mapping function depends merely on MCS.

The mapping function is shown as (1).

$$SNR_{eff} = \beta f^{-1} \left(\frac{1}{R} \sum_{r=1}^R f \left(\frac{SNR_r}{\beta} \right) \right) \quad (1)$$

where, $f(x)$ is a compression function of the effective SNR mapping, $f^{-1}(x)$ is an inverse function of $f(x)$, R denotes the RB's number, and β is an adjustment factor, which can be adjusted to match the MCS. According to the compression function used, the effective SNR mapping algorithm can be divided into exponential effective SNR mapping(EESM) and mutual information effective SNR

mapping(MIESM) algorithm. Among them, MIESM needs to calculate the channel capacity without bit interleaving coding and modulation, and its calculation amount is much larger than that of EESM.

C. Optimization of adjustment factor

In practical engineering applications, the UE calculates the SNR of the received signal, uses the effective signal-to-noise ratio mapping algorithm to obtain the effective SNR in the AWGN environment, and then obtains the CQI measurement value through the relationship between CQI and SNR. The accuracy of CQI measurement is related to the SNR estimation error and the accuracy of the adjustment factor in the effective SNR mapping algorithm. The SNR estimation error is determined by the channel estimation. Therefore, this paper mainly corrects the adjustment factor to improve the accuracy of the effective SNR mapping algorithm.

The optimization of adjustment factor is a gradual process. An iterative algorithm to determine the best factor value according to the convexity of the mean square error function can accurately obtain the factor value under a certain channel model. In order to make the mapping process of effective signal-to-noise ratio more accurate, another adjustment factor is introduced in this paper. The expression of the SNR mapping algorithm after adding the adjustment factor is as follows.

$$SNR_{eff} = \alpha f^{-1} \left(\frac{1}{R} \sum_{r=1}^R f \left(\frac{SNR_r}{\beta} \right) \right) \quad (2)$$

The purpose of the correction adjustment factor is to make the predicted BLER under AWGN as close to the actual simulated BLER as possible. Generally, the minimum mean square error method is used for calibration. Since the magnitude of BLER is generally at least as low as 1e-3, adding and subtracting to BLER and then squaring on this basis may cause too many digits after the decimal point and reduce accuracy. In response to this problem, this paper proposes a scheme that takes the BLER value to logarithm and then performs MSE processing to find the optimal value. In addition, in the calibration process, in order to reflect the channel situation as much as possible, this article uses multiple channels to achieve, and averages the BLER values on multiple channels to calculate the MSE. The MSE calculation formula used in this paper is given by (3).

$$MSE((\alpha, \beta)_b) = \sum_{j=1}^{N_s} \left[\log_{10}(BLER_{LUT}(\alpha, \beta)_b) - \log_{10}(BLER) \right]^2 \quad (3)$$

The optimization algorithm of adjustment factors proposed in this paper can be concluded as Algorithm 1 in Table III. Where $(\alpha, \beta)_b, b=1, \dots, N_{ab}$ denotes b th adjustment factor, N_{ab} denotes the numbers of candidate factors. N_c is the number of channel realizations, which should be large enough to show characters of whole channel. N_s means number of SNR vectors. The noise variance calculated by the receiver corresponding to each SNR is $\sigma_j^2, j=1, \dots, N_s$. $BLER_{LUT}(:, :)$ is BLER obtained by mapping $SNR_{eff}(:, :)$ under AWGN curve, while $BLER(:, :)$ means BLER calculated by actual simulation.

TABLE III. OPTIMIZATION ALGORITHM OF ADJUSTMENT FACTORS

Algorithm 1 optimization algorithm of adjustment factors	
Input: $(\alpha, \beta) = \{(\alpha, \beta)_1, (\alpha, \beta)_2, \dots, (\alpha, \beta)_{N_{ab}}\}$	
1: for $b = 1$ to N_{ab}	
2: for $k = 1$ to N_c channel realization	
3: for $j = 1$ to N_s SNR	
4: $BLER = 0$	
5: for $l = 1$ to 1000 frame	
6: generate channel matrix $\mathbf{H}_k$	
7: calculate noise σ_j^2 , $SNR_{eff}(\mathbf{H}_k, \sigma_j^2, (\alpha, \beta)_b)$ and error blocks	
8: end	
9: calculate $BLER(\mathbf{H}_k, \sigma_j^2)$	
$BLER_{LUT}(\mathbf{H}_k, \sigma_j^2, (\alpha, \beta)_b) = BLER_{AWGN}(SNR_{eff}(\mathbf{H}_k, \sigma_j^2, (\alpha, \beta)_b))$	
10: end	
11: end	
12: calculate average BLER by N_c	
$BLER(\sigma_j^2) = \frac{1}{N_c} \sum_{k=1}^{N_c} BLER(\mathbf{H}_k, \sigma_j^2)$	
$BLER_{LUT}(\sigma_j^2, (\alpha, \beta)_b) = \frac{1}{N_c} \sum_{k=1}^{N_c} BLER_{LUT}(\mathbf{H}_k, \sigma_j^2, (\alpha, \beta)_b)$	
13: calculate MSE	
$MSE((\alpha, \beta)_b) = \sum_{j=1}^{N_s} [\log_{10}(BLER_{LUT}(\sigma_j^2, (\alpha, \beta)_b)) - \log_{10}(BLER(\sigma_j^2))]^2$	
14: end	
Output: $(\tilde{\alpha}, \tilde{\beta}) = \arg \min_{(\alpha, \beta)} MSE(\alpha, \beta)_b$	

IV. EXPERIMENTAL RESULTS

This paper first simulates the modulation and coding schemes in Table I, Table II on a Gaussian channel based on 5G PDSCH link to obtain the SNR-BLER curve of each MCS, and finds the corresponding SNR threshold when $BLER = 0.1$. Then we simulate in the 5G channel model to get the SNR-BLER curve. Finally, the adjustment factor is optimized to obtain the optimal value of the adjustment factor under each MCS. The configuration is as follows:

TABLE IV. SIMULATION PARAMETERS

Parameters	Configuration
Channel type	AWGN, CDL-A
Modulation type	QPSK, 16QAM, 64QAM, 256QAM
SNR	-10:0.3:26
Code rate	Refer Table I,II
Number of cycles	2000

A. CQI-SNR mapping under AWGN

In this paper, the SNR-BLER curves of 15 MCS simulations under AWGN SISO system by 20,000 frames in three CQI tables are shown in the figure. The mapping relationship between CQI and SNR in Table I obtained by straight line fitting is shown as Fig.1.

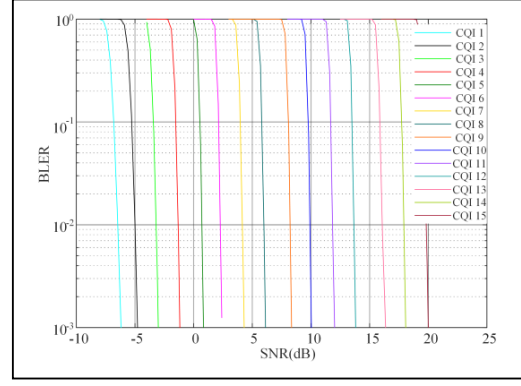


Fig. 1. SNR-BLER curve in Table I.

From Fig.1, the SNR threshold of CQI in Table I is [-6.82, -5.28, -3.44, -1.53, -0.53, 2.12, 3.96, 5.80, 8.08, 9.76, 11.72, 13.49, 15.87, 17.73, 19.63]. The curve can be fitted as Fig2. The linear relationship expression is as (4).

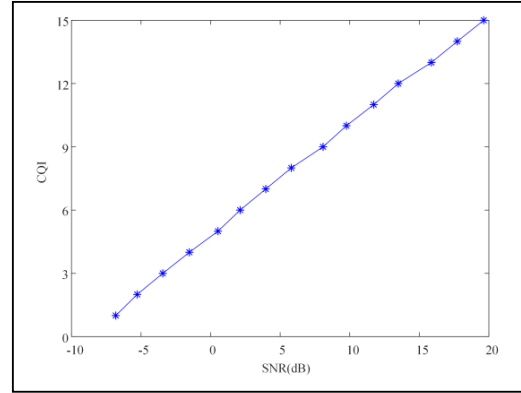


Fig. 2. SNR-CQI curve in Table I.

$$CQI = \begin{cases} 0.5633 * SNR + 4.9040 & SNR < 0 \\ 0.5639 * SNR + 4.7443 & 0 \leq SNR < 7 \\ 0.5493 * SNR + 4.5880 & 7 \leq SNR < 14 \\ 0.5319 * SNR + 4.5624 & SNR \geq 14 \end{cases} \quad (4)$$

For the SNR corresponding to the 15 MCS in Table II, the SNR-BLER curves is shown as Fig3

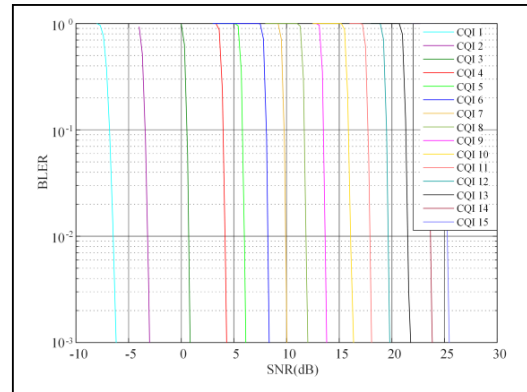


Fig. 3. SNR-BLER curve in Table II.

From Fig.3, the SNR threshold of CQI in Table 1 is [-6.82,-3.44,-0.53,3.79,5.80,8.08,9.76,11.72,13.49,15.87,17.73,19.50,21.32,23.51].The curve can be fitted as Fig4. The linear relationship expression can be expressed as (5).

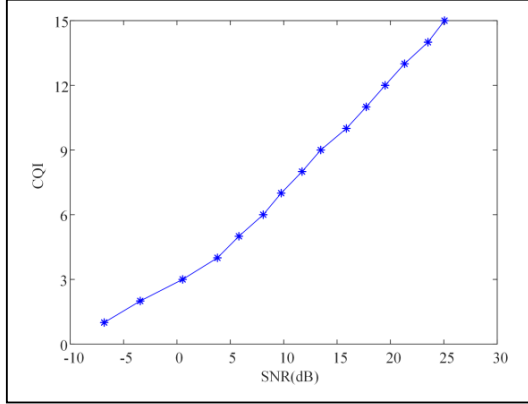


Fig. 4. SNR-CQI curve in Table II..

$$CQI = \begin{cases} 0.2789 * SNR + 2.9142 & SNR < 4 \\ 0.5246 * SNR + 1.8749 & 4 \leq SNR < 14 \\ 0.5369 * SNR + 1.4946 & SNR \geq 14 \end{cases} \quad (5)$$

B. Determination of Optimational Adjustment Factor

This section simulates the channel model CDL-A defined in the 5G standard to determine the optimal adjustment factor. In the simulation process, first obtain different channel implementations in the CDL-A channel; then calculate the existing subcarrier SNR value for each channel implementation, then calculate the effective SNR through the EESM algorithm, and finally obtain the BLER MSE to determine The optimal adjustment factor value.

According to the above steps, the adjustment factor values corresponding to CQI 1-15 in Table I and Table II are sequentially obtained as shown in Table V and Table VI.

TABLE V. OPTIMAL ADJUSTMENT FACTORS TABLE I

CQI index	α	β
1	3.02	2.96
2	2.02	2.02
3	1.81	1.79
4	1.17	1.15
5	1.42	1.41
6	1.36	1.35
7	3.55	3.63
8	3.60	3.56
9	5.60	5.57
10	16.19	16.17
11	13.49	13.45
12	17.77	17.85
13	24.04	24.11
14	29.34	29.23

CQI index	α	β
15	31.16	31.29

TABLE VI. OPTIMAL ADJUSTMENT FACTOR TABLE 2

CQI index	α	β
1	3.02	2.96
2	1.81	1.79
3	1.42	1.41
4	3.55	3.63
5	3.60	3.56
6	5.60	5.57
7	16.19	16.17
8	13.49	13.45
9	17.77	17.85
10	24.04	24.11
11	29.34	29.23
12	40.23	40.05
13	42.33	42.55
14	46.78	47.03
15	49.89	49.59

The BLER curve after the effective SNR mapping of CQI 1-15 in the CDL-A channel through the AWGN channel is shown as Fig.5,6.

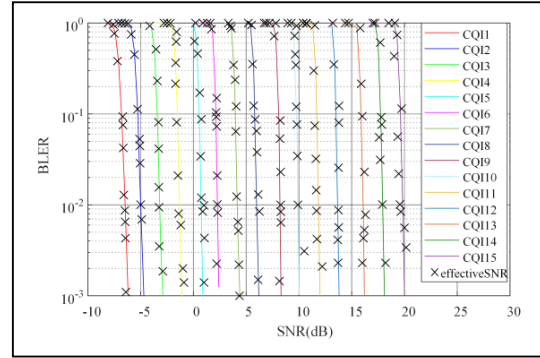


Fig. 5. Effective SNR mapping curve under CDL-A channel in Table I..

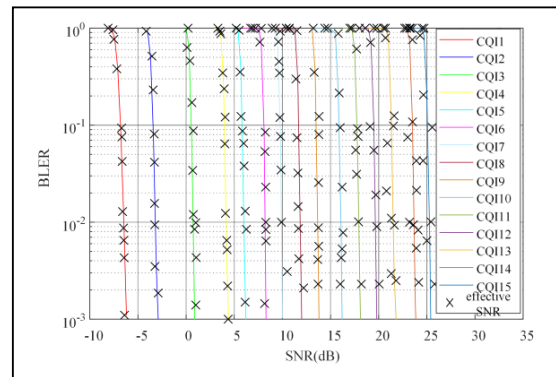


Fig. 6. Effective SNR mapping curve under CDL-A channel in Table II..

Fig.5 and 6 are the simulated effective SNR mapping curves of CQI in Table I and II under 5G CDL-A channel. In Fig.5 and Fig.6, the solid line represents the SNR-BLER curve under the AWGN channel, and the black discrete mark represents the BLER curve simulated and mapped in the CDL-A channel. The adjustment factors used in the effective SNR mapping process are shown in the table V and VI. It can be seen from the figure that the discrete marker points are relatively concentrated on the AWGN curve, which proves that the adjustment factor correction algorithm used in this paper can complete the EESM mapping well.

V. CONCLUSION

The development of IoT will bring vitality to innovative applications in the field of Internet of Things, and bring innovative breakthroughs in many fields such as smart cities. The IoT-5G joint technology will promote this process. AMC technology has a crucial impact on the performance of 5G mobile communication systems. This paper introduces and studies the adaptive coding and modulation technology in the downlink of the physical layer of the 5G system. The SNR-BLER curves corresponding to the CQI tables in TS38.214 are obtained through the new AWGN simulation. Then, the SNR thresholds corresponding to different modulation and coding schemes in each table are obtained, and the relationship expression between SNR and CQI is obtained by a straight line fitting method. Finally, an exponential effective SNR mapping algorithm with improved adjustment factor optimization is proposed. The simulation proves that the performance of the improved adjustment factor optimization algorithm used in this paper is significantly improved.

REFERENCES

- [1] Ammar M, Russello G, Crispo B. Internet of Things: A survey on the security of IoT frameworks[J]. *Journal of Information Security and Applications*, 2018, 38: 8-27
- [2] Amin A A , Basak D , Khadem T , et al. Analysis of modulation and coding scheme for 5th generation wireless communication system[C]// *International Conference on Computing*. IEEE, 2017.
- [3] Ali M A, Jasmin E A. Physical layer improvements in 4G/5G wireless communication networks through link adaptation techniques[C]// *International Conference on Power, Instrumentation, Control and Computing*. IEEE, 2016:1-5
- [4] Series, M. IMT Vision-Framework and overall objectives of the future development of IMT for 2020 and beyond[J] *Recommendation ITU*, 2015, 2083-0
- [5] Lee H, Kim H, Park H. Novel Calibration of MIESM and Reduction of CQI Feedback for Improved Fast Link Adaptation[J]. *Electronics*, 2019, 8(3): 278
- [6] Qadri M I, Zia M. Adaptive modulation and coding with selective retransmission under OFDM signaling[J]. *Wireless Personal Communications*, 2018, 101(4): 1787-1805
- [7] Latha T, Priyadharsini S. Effective communication in LTE-A using link adaptation technique[C]// *International Conference on Innovations in Information*. IEEE, 2015
- [8] Ahmad I, Chang K. Effective SNR Mapping and Link Adaptation Strategy for Next-Generation Underwater Acoustic Communications Networks: A Cross-Layer Approach[J]. *IEEE Access*, 2019, 7: 44150-44164.
- [9] Xue R, Yu H, Cheng Q. Adaptive Coded Modulation Based on Continuous Phase Modulation for Inter-Satellite Links of Global Navigation Satellite Systems[J]. *IEEE Access*, 2018, 6: 20652-20662
- [10] Ryoo S, Choi K, Lee K, et al. Performance Improvement of the Combined AMC-MIMO Systems with MCS Level Selection Technique[M]// *Dynamic Balancing of Mechanisms and Synthesizing of Parallel Robots*. 2015:3-3
- [11] Alnwaimi G, Boujemaa H. Adaptive packet length and MCS using average or instantaneous SNR[J]. *IEEE Transactions on Vehicular Technology*, 2018, 67(11): 10519-10527
- [12] 3GPP TS 38.214, Physical layer procedures for data V15.2.0 .2018
- [13] Abdulhsan, Muntadher Q , Salman M I , NG C K , et al. Approximate linear minimum mean square error estimation based on channel quality indicator feedback in LTE systems[C]//*IEEE Malaysia International Conference on Communications*, No vember 26-28, 2013, Kuala Lumpur,Malaysia. New Jersey:IEEE Press, 2013 : 446-451